SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Hari Charan.P (192511107) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Hari Charan.P

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Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

   a. Deduce the most likely original message.
   b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

   a. Determine the most probable original message by reversing the encryption process.
   b. Critically evaluate why such a method is vulnerable in modern communication systems.
   c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 20	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 25	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 5	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CO-1-AT-I

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Scenario: Event Communication Security

A Student organization wants to securely communicate confidential event details using a simple classical encryption technique. The given msg is "MEET AT NOON" and a fixed no. of position in the alphabet.

Encrypted message using shift of 4

This method is called the Caesar cipher. In Caesar Cipher, each letter is shifted by a fixed no. of positions in the Alphabet.

For a Shift 4 Positions forward:

Original M E E T A T N O O N
+4 shift Q I I X E X R S S R

Therefore:

Original message:

MEET AT NOON

Encrypted message:

QIIX EX RSSR

Spaces are kept uncharged during Encrypt

b) Decryption by an authorized member,

An authorized member who knows the encryption rule can recover the original message by shifting every letter 4 positions backward. This process is called decryption.

For eg:-

- ✱ Q → M ∴ QIIX EX RSSR
- ✱ I → E ↓ Shift each letter 4 Position backward
- ✱ I → E
- ✱ * → T MEET AT NOON
- ✱ E → A Thus, the authorized member can successfully retrieve the original confidential message.
- ✱ T → *
- ✱ R → N
- O → S
- ~~O~~ O → S
- R → N

Conclusion:-

The Caesar Cipher is a simple substitution encryption technique. It is easy to implement and understand, but because it has only a limited no. of possible shifts, it is not considered secure for modern confidential communication.

Scenario: Intercepted Suspicious Message

A Cybersecurity intern intercepts the coded message

GSRH RH Z HVXIVG

The Structure suggest that a simple substitution Cipher has been used.

Most likely original message

The most likely encryption method is Atbash Cipher.

In Atbash cipher, the alphabet is reversed:

Plain A B C D E F G H I J

Cipher Z Y X W V U T S R Q

Similarly:

Plain K L M N O P Q R S T

Cipher P O N M L K J I H G

Applying the reverse substitution:

GSRH RHZHVXIVG

becomes:

This is A SECRET

Therefore, the original msg is,

"THIS IS A SECRET"

b) Justification

The answer can be justified using the following observations:

1. GSRH contains four letters and follows the same pattern as THIS.
2. RH converts to IS.
3. Z converts to A.
4. HVXIVG converts to SECRET
5. The substitution follows the Atbash rule, where the alphabet is reversed.
6. Eg:
   - G → T
   - S → H
   - R → I
   - H → S
   - Z → A
   - V → E
   - X → C
   - I → R
   - G → T

Therefore, GSRH RH HVXIVG
↓ Atbash decryption
↓ This is A secret

The intercepted message is most likely encrypted using the Atbash substitution cipher. The repeating word patterns and consistent reverse alphabet substitutions confirm that the original message is "THIS IS A SECRET".

Pattern analysis :-

* GSRH has four characters
* RH has two characters and maps to IS.
* Z maps to A.
* HVXIVG has six characters and maps to.

SECRET :-

* The resulting sentence is grammatically meaningful.

Result :-

This is a secret.

Analysis :-

Atlash is a simple ~~Caesar~~ substitution method. Since the substitution pattern is fixed, it does not provide strong modern security.

The interpted message is most likely "This is a secret".

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